MAHANDAR KUMAR

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Portfolio | GitHub | LinkedIn | Codeforces | LeetCode

SUMMARY

Computer Science undergraduate at FAST-NUCES Karachi (CGPA 3.95) who builds RAG chatbots and full-stack web applications with Python, LangChain, Node.js, and Next.js. AWS Certified Generative AI Developer with a completed 2-month AI internship at FlyRank and an active competitive programming record on Codeforces and LeetCode. Seeking a software engineering or AI engineering internship.

EDUCATION

Bachelor of Science in Computer Science **Expected 2029**
FAST National University of Computer & Emerging Sciences (FAST-NUCES), Karachi *CGPA: 3.95*

- Coursework: Data Structures & Algorithms, Object-Oriented Programming, AI/ML fundamentals.

EXPERIENCE

AI Intern, FlyRank Remote | 2 months (completed)

- Completed a 2-month remote internship focused on AI fluency, applying AI tools and prompt-based workflows to real project tasks.

Volunteer, ILM KI ROSHNI Community Engagement Project Community project

- Taught underprivileged students through a volunteer initiative and wrote the project report and proposal documentation.

PROJECTS

WhatsApp RAG Chatbot with Personal-Fact Memory GitHub: github.com/mk-2007/WhatsApp-Bot---AI-Noor-Institute
Node.js (Baileys), Python (LangChain), Groq API, ChromaDB, FAISS

- Built a WhatsApp chatbot with a Node.js messaging layer and a Python RAG backend that answers from a custom knowledge base stored in a persistent on-disk vector store.
- Added automatic extraction and recall of user facts (name, location, occupation, interests) so replies stay personalized across a conversation.

Multimodal RAG System from Scratch GitHub: github.com/mk-2007/Multimodal-Rag-ChatBot
Python, Vector Embeddings, Retrieval-Augmented Generation

- Implemented a RAG pipeline from first principles: ingests research-paper text, generates embeddings, and retrieves relevant passages to ground LLM answers.

RAG AI Chatbot (Text-Based) GitHub: github.com/mk-2007/AI-Season---RAG-ChatBot
LangChain, Hugging Face, ChromaDB, ChatGroq (Llama 3.1 8B), Streamlit

- Built a retrieval chatbot using Hugging Face sentence-transformer embeddings, an in-memory ChromaDB vector store, and Llama 3.1 8B served through the Groq API.
- Developed a custom Streamlit interface for the chat experience.

TypeMaster: Console Typing Speed Game GitHub: github.com/mk-2007/TypeMaster---C-typing-appilcation
C++ (OOP)

- Built a console typing speed game in C++, using inheritance, polymorphism, and encapsulation to structure gameplay logic and scoring.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL
AI / LLM: Retrieval-Augmented Generation (RAG), LLM applications, LangChain, prompt engineering, ChromaDB, FAISS, vector embeddings, Hugging Face, Groq API, Ollama
Web & Backend: Node.js, Express, FastAPI, REST APIs, SQLite
Frontend: React, Next.js, React Three Fiber, Three.js, Streamlit
Tools: Git, GitHub, Vercel
Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, competitive programming (greedy, simulation, cycle detection, constructive algorithms)

CERTIFICATIONS

- AWS Certified Generative AI Developer, Amazon Web Services